REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 1 Lesson 2 | Mathematics and Physics

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: v = d/t; ρ = m/V; A = length × width; 1 km = 1000 m; 1 mm = 10^{-3} m.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 9–12.

P01. Which is an SI base quantity?

A. Speed B. Area
C. Time D. Density

Answer

C. Time.

P02. Which prefix means one thousandth?

A. Kilo B. Mega
C. Giga D. Milli

Answer

D. Milli.

P03. How many significant figures are in 0.00450?

A. Three B. One
C. Two D. Five

Answer

A. Three.

P04. Which unit is appropriate for speed in SI?

A. m s B. m/s
C. kg/m D. s/m

Answer

B. m/s.

P05. Why must both sides of a physical equation have the same dimensions?

A. They must have the same numerical value in every unit system B. All physical quantities have identical dimensions
C. They represent the same kind of physical quantity D. Units can be ignored

Answer

C. They represent the same kind of physical quantity.

P06. Convert 5 mm to metres.

A. 0.005 m B. 0.01 m
C. 0.0025 m D. -0.005 m

Answer

A. 0.005 m. 1 mm = 10^{-3} m, so 5 mm = 5×10^{-3} = 0.005 m.

P07. Convert 2.4 km to metres.

A. 4800 m B. 2400 m
C. 1200 m D. -2400 m

Answer

B. 2400 m. 1 km = 1000 m, so $2.4 \times 1000 = 2400$ m.

P08. A car travels 240 m in 7 s. Calculate its average speed.

A. 68.57 m/s B. 17.14 m/s
C. 34.29 m/s D. -34.29 m/s

Answer

C. 34.29 m/s. Average speed = distance/time = 240/7 = 34.29 m/s.

P09. A sample has mass 160 g and volume 20 cm^3. Calculate its density in g/cm^3.

A. 16 g/cm^3 B. 4 g/cm^3
C. -8 g/cm^3 D. 8 g/cm^3

Answer

D. 8 g/cm^3. $\rho = m/V = 160/20 = 8$ g/cm^3.

P10. A rectangle is 4 cm long and 3 cm wide. Calculate its area.

A. 12 cm^2 B. 24 cm^2
C. 6 cm^2 D. -12 cm^2

Answer

A. 12 cm^2. A = length × width = $4 \times 3 = 12$ cm^2.

P11. What is dimensional analysis?

Answer

It checks the consistency of equations or unit conversions by comparing the physical dimensions of quantities.

P12. Distinguish a base quantity from a derived quantity.

Answer

A base quantity is defined independently in the SI system; a derived quantity combines base quantities, such as speed = length/time.

P13. Explain the meaning of the prefix kilo.

Answer

Kilo denotes multiplication by 10^3, or 1000.

P14. Explain why significant figures are important.

Answer

They communicate the precision supported by measured data and prevent results from implying unjustified precision.

P15. State the rounding rule for multiplying measured quantities.

Answer

Round the final result to the same number of significant figures as the measured factor with the fewest significant figures.

P16. Convert 7.5 mm to metres.

Answer

1 mm = 10^{-3} m, so 7.5 mm = 7.5×10^{-3} = 0.0075 m.

P17. Convert 10 mm to metres.

Answer

1 mm = 10^{-3} m, so 10 mm = 10×10^{-3} = 0.01 m.

P18. Convert 3.6 km to metres.

Answer

1 km = 1000 m, so $3.6 \times 1000 = 3600$ m.

P19. Convert 4.8 km to metres.

Answer

1 km = 1000 m, so $4.8 \times 1000 = 4800$ m.

P20. A car travels 360 m in 8 s. Calculate its average speed.

Answer

Average speed = distance/time = 360/8 = 45 m/s.

P21. A car travels 480 m in 9 s. Calculate its average speed.

Answer

Average speed = distance/time = 480/9 = 53.33 m/s.

P22. A sample has mass 240 g and volume 30 cm^3. Calculate its density in g/cm^3.

Answer

$\rho = m/V = 240/30 = 8$ g/cm^3.

P23. A sample has mass 320 g and volume 40 cm^3. Calculate its density in g/cm^3.

Answer

$\rho = m/V = 320/40 = 8$ g/cm^3.

P24. A rectangle is 6 cm long and 4 cm wide. Calculate its area.

Answer

A = length × width = $6 \times 4 = 24$ cm^2.

P25. A rectangle is 8 cm long and 5 cm wide. Calculate its area.

Answer

A = length × width = 8 × 5 = 40 cm^2.

P26. Check whether d = vt is dimensionally consistent when d is distance, v is speed and t is time.

Answer

Yes. [v][t] = (length/time)(time) = length = [d].

P27. A student writes speed = distance × time. Identify the problem using units.

Answer

The units would be m s, not m/s. Speed requires distance divided by elapsed time.

P28. Calculate 2.4 m × 3.12 m and report the correct significant figures.

Answer

The product is 7.488 m^2. The factor 2.4 has two significant figures, so report 7.5 m^2.

P29. Calculate 12.35 cm + 0.4 cm with appropriate precision.

Answer

The unrounded sum is 12.75 cm. For addition, retain the least precise decimal place: 12.8 cm.

P30. An answer is numerically plausible but has units kg/s for a speed. Can it be accepted? Explain.

Answer

No. Speed has dimensions of length/time and units m/s. Correct dimensions are a necessary check even when a number appears plausible.